

Project Proposal

Vision-Guided Autonomous Material Handling Robot

Guided-Navigation Fork-Lifter with Image-Processing-Based Product Classification

Project Type: Mechatronics / Robotics / Automation Prototype

Navigation Method: IR-based guided path tracking

Decision Method: Camera-based color classification using OpenCV

Main Controllers: Raspberry Pi 4B and Arduino Mega 2560

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Course / Project: ME 4408 - MIC Lab

1. Abstract

Modern delivery areas, small warehouses, laboratories, and production units often require repeated movement of lightweight products from a loading point to categorized docking stations. Manual handling is slow, repetitive, and prone to human error. This project will develop a compact autonomous material handling robot that will demonstrate guided navigation, product identification, autonomous destination selection, and fork-lifter-based product handling.

The robot will follow a prepared black-line route using an IR line sensor array. At the pickup station, the product image will be captured using a webcam or Pi camera. A Raspberry Pi will process the image using OpenCV and classify the product color. Based on the detected color, the robot will select the correct docking station and carry the product using a compact fork-lifter mechanism.

The system will use Raspberry Pi 4B for image processing and high-level mission decisions. Arduino Mega 2560 will handle line following, motor control, ultrasonic sensing, and servo actuation. Four 25GA gear motors will provide stable movement, while an MG996R servo will operate the fork-lifter mechanism. The final prototype will be expected to pick, classify, transport, and place lightweight products into correct docking stations.

2. Objectives

The main objective of this project will be to design and build a vision-guided autonomous material handling robot for lightweight product transportation.

The specific objectives are:

- A guided-navigation robot will be developed using an IR line sensor array.
- Image-processing-based product color classification will be implemented using Raspberry Pi and OpenCV.
- A compact fork-lifter mechanism will be designed for lightweight objects.
- Arduino Mega will be used for real-time control of motors, sensors, and actuators.
- The robot will select the correct docking station based on detected product color.
- Ultrasonic distance sensing will be used for obstacle and docking safety.
- A stable power system will be designed using regulated supply rails.
- The complete system will be documented for reproducibility and future improvement.

3. Introduction / Background

An autonomous material handling robot is a simplified form of an automated guided vehicle used in logistics, warehouses, laboratories, and production systems. In industrial applications, similar robots are used to transport goods between loading docks, storage zones, and processing stations.

For this prototype, a guided black-line path will be used for reliable navigation. This path will act as a structured route, similar to guided paths used in automated guided vehicles. The main autonomous decision-making will be performed by the vision system. The robot will use image processing to identify the product color and choose the correct docking station.

Instead of using only a simple color sensor, a camera-based image processing system will be used. The captured image will be converted to HSV color space, and color masks will be used to detect red, blue, and green products. The detected color will be mapped to the corresponding docking station.

The fork-lifter mechanism will allow the robot to perform a material-handling task instead of only moving like a normal line-following robot. The robot will lift a small product, carry it along the guided route, and place it into the selected docking station.

4. Methodology

The project will follow a mechatronic development process: requirement definition, sub-system design, component selection, fabrication, integration, calibration, testing, and improvement.

The major development steps will be:

1. Product size, maximum payload, route layout, color categories, docking stations, and safety limits will be defined.
2. A lightweight chassis will be designed with motor mounts, camera mount, line sensor bracket, and fork-lifter mechanism.
3. The IR line sensor array, camera, ultrasonic sensors, and optional limit switches will be integrated.
4. Line-following control will be implemented using Arduino Mega.
5. Color classification will be implemented using Raspberry Pi and OpenCV.
6. Serial communication will be established between Raspberry Pi and Arduino Mega.
7. Pickup, classification, route selection, delivery, and return operation will be tested.

4.1 Core Algorithm

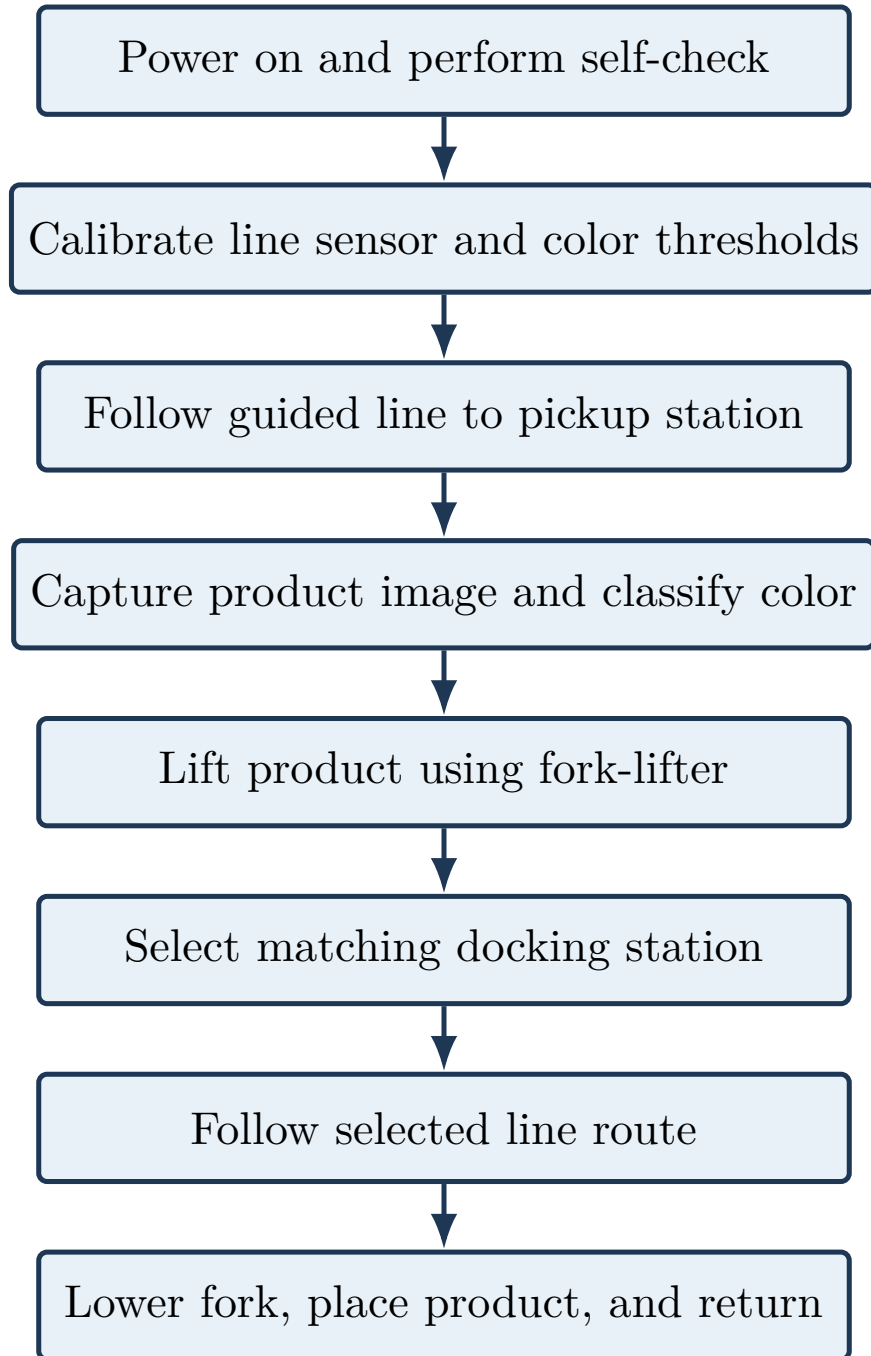
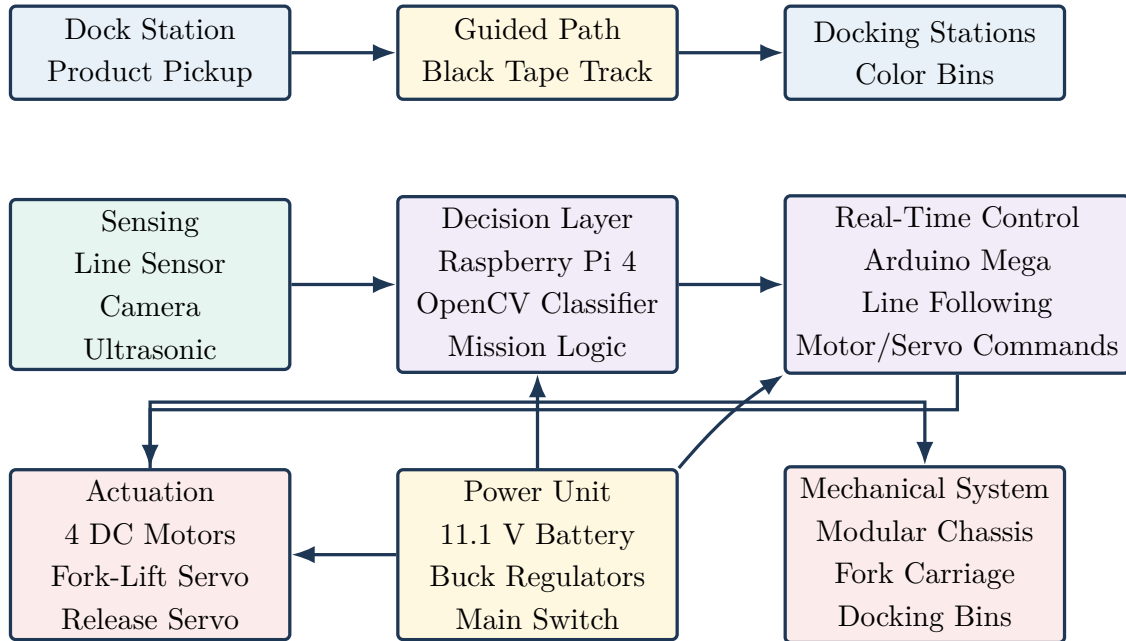


Figure 1: Core algorithm of the proposed robot.

5. System Design and Architecture

The system will be divided into sensing, decision-making, real-time control, actuation, mechanical structure, and power management. Raspberry Pi will handle vision and mission decisions, while Arduino Mega will handle time-critical motor and actuator control.

5.1 System Architecture



Closed-loop operation: sense $\rightarrow$ classify $\rightarrow$ decide $\rightarrow$ move/lift/drop $\rightarrow$ verify $\rightarrow$ return

Figure 2: Proposed system architecture for the autonomous material handling robot.

5.2 Control Architecture

- **Low-level loop:** Arduino Mega will read the line sensor array, compute line error, and control motor speed.
- **High-level loop:** Raspberry Pi will capture the product image, classify color, select destination, and send mission commands.
- **Safety loop:** Ultrasonic sensors will stop the robot if an obstacle is detected within the safety distance.
- **Product-handling loop:** The fork-lifter servo will lift, carry, and place the product according to the mission state.

6. Operational Flowchart

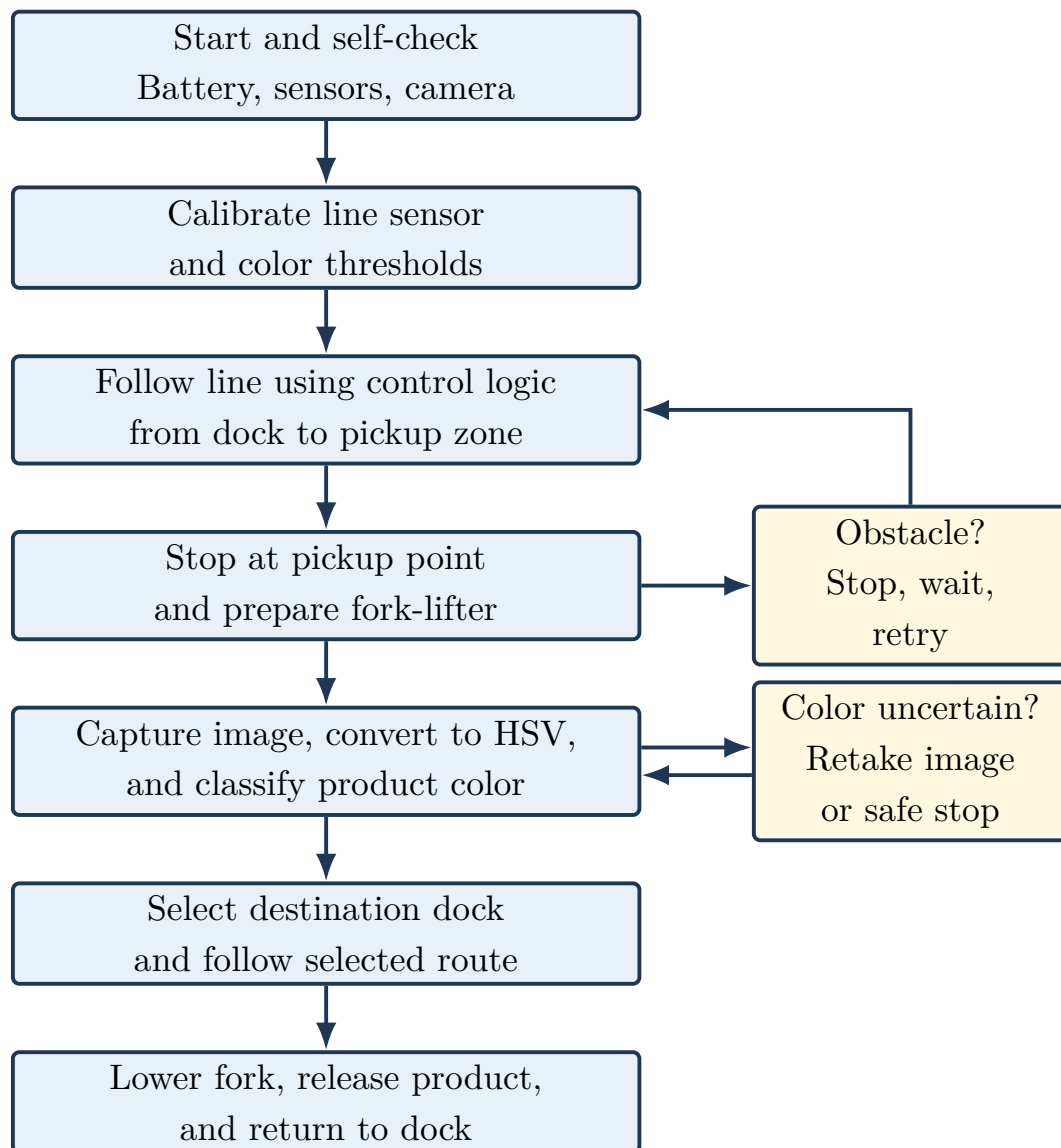
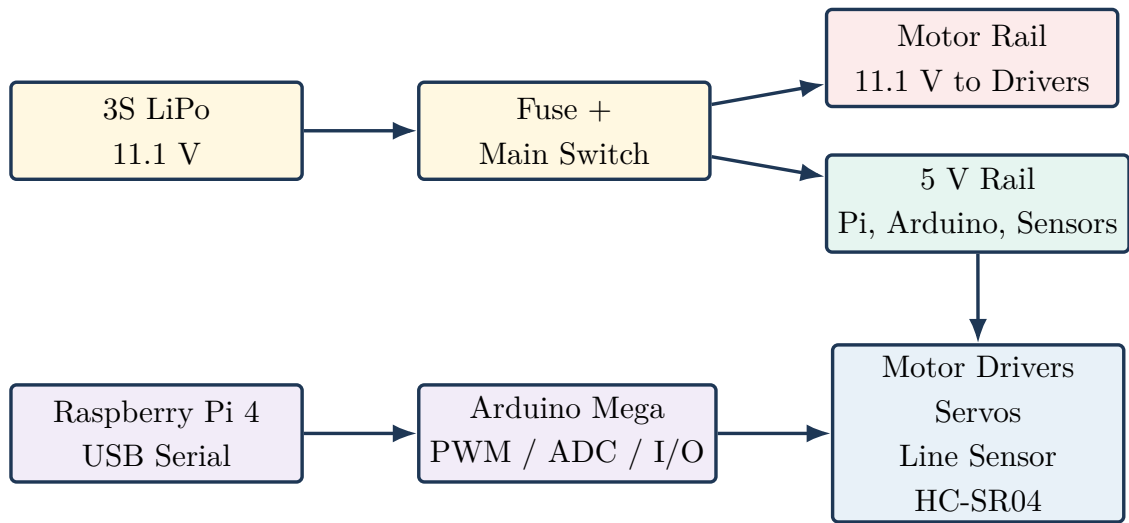


Figure 3: Operational flowchart showing the autonomous decision cycle.

7. Simplified Power and Signal Distribution



Ground will be common; high-current motor wiring will be routed away from camera and sensor lines.

Figure 4: Simplified power and signal distribution plan.

8. Suggested Pin Allocation

Subsystem	Connection	Purpose
Line sensor array	Arduino analog/digital pins	Line position measurement
Left motor driver	Arduino PWM/DIR pins	Left side wheel control
Right motor driver	Arduino PWM/DIR pins	Right side wheel control
Fork-lifter servo	Arduino PWM pin	Product lifting and lowering
Release servo	Arduino PWM pin	Optional release action
Ultrasonic sensor	Arduino trigger/echo pins	Obstacle and docking detection
Camera	Raspberry Pi USB/CSI port	Product image acquisition
Pi-Arduino link	USB serial	Mission commands and status messages

9. Bill of Materials and Cost Estimation

The cost estimation is prepared in Bangladeshi Taka. Prices may vary depending on availability and final material choice.

No.	Component	Qty	Unit Price	Total	Purpose
1	Raspberry Pi 4B	1	6,000	6,000	Vision processing and mission logic
2	USB webcam / Pi Camera V2	1	1,150–2,300	1,150–2,300	Product image capture
3	Arduino Mega 2560 CH340	1	2,250	2,250	Real-time control
4	5-channel IR line sensor array	1	600–800	600–800	Guided path tracking
5	25GA 12V 130RPM gear motor	4	750	3,000	Four-wheel drive
6	L298N motor driver	2	185	370	Motor control
7	MG996R servo	1	730	730	Fork-lifter mechanism
8	SG90 servo	1	129–170	129–170	Optional release mechanism
9	HC-SR04 ultrasonic sensor	2	95–150	190–300	Obstacle and docking detection
10	LM2596 buck converter	2	490	980	Regulated power rails
11	3S 11.1V LiPo battery	1	2,500	2,500	Main power source
12	LiPo balance charger	1	500–800	500–800	Battery charging
13	Wheels for 25GA motors	4	100–180	400–720	Robot movement
14	Chassis / frame material	1 set	1,000–1,500	1,000–1,500	Mechanical body
15	Fork-lifter materials	1 set	1,000–1,500	1,000–1,500	Fork structure
16	Wires, connectors, switches, screws, mounts	1 set	800–1,000	800–1,000	Assembly and wiring
17	Black tape, docking sheets, bins, sample products	1 set	500–700	500–700	Test arena
Estimated Minimum Total				22,199	
Estimated Comfortable Total				25,400	

Table 1: Bill of materials and cost estimation.

10. Component and Technical Details

Component	Technical Details	Role in Project
Raspberry Pi 4B	Linux-based single-board computer	Vision processing and mission logic
Camera / Webcam	Product image capture device	Color classification
Arduino Mega	Multiple I/O pins and PWM outputs	Motor, sensor, and servo control
IR line sensor array	Detects black line position	Guided navigation
25GA motors	12V, 130RPM gear motors	Robot locomotion
L298N drivers	Dual H-bridge modules	Motor driving
MG996R servo	High-torque servo	Fork-lifter operation
HC-SR04	Ultrasonic distance sensor	Obstacle and dock detection
LM2596	DC-DC buck converter	Voltage regulation

11. Software Modules

11.1 Raspberry Pi Modules

- **camera_capture.py:** This module will capture product images.
- **color_classifier.py:** This module will classify red, blue, and green products.
- **mission_manager.py:** This module will select the correct docking station.
- **serial_communication.py:** This module will communicate with Arduino Mega.

11.2 Arduino Modules

- **line_following.ino:** This program will perform guided route tracking.
- **motor_control.ino:** This program will control motor speed and direction.
- **forklift_control.ino:** This program will control the fork-lifter servo.
- **safety_monitor.ino:** This program will handle ultrasonic-based safety response.

12. Gantt Chart: 10-Week Timeline

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Requirement finalization and design	X	X								
Component purchase and inspection	X	X								
Mechanical chassis and fork fabrication		X	X	X						
Power distribution and wiring			X	X						
Line following calibration			X	X	X					
Camera setup and color classification				X	X	X				
Fork pickup and drop integration					X	X				
Route logic and serial communication						X	X			
Reliability testing and tuning								X	X	
Final documentation and presentation									X	X

13. Expected Outcomes / Final Goals

The expected final outcomes are:

- The robot will follow the guided black-line route reliably.
- The robot will stop at the pickup station.
- At least three colors will be classified: red, blue, and green.
- The fork-lifter will lift and carry a lightweight product.
- The correct docking station will be selected based on product color.
- The product will be placed at the correct station and the robot will return.
- The complete system will demonstrate sensing, decision-making, actuation, and closed-loop control.

14. Applications

- Educational AGV and robotics demonstration.
- Small warehouse sorting model.
- Laboratory sample delivery system.
- Factory-line automation teaching model.
- Color-coded inventory transportation.
- Prototype platform for future smart logistics research.

15. Challenges and Risk Mitigation

Challenge	Possible Effect	Mitigation Strategy
Lighting variation	Wrong color classification	Fixed lighting and HSV calibration will be used.
Line sensor error	Robot may leave the path	Sensor height and line-following gain will be tuned.
Fork misalignment	Product may fail to lift	Fixed pickup position and guide rails will be used.
Voltage drop	Controller may reset	Separate regulated power rails will be used.
Motor driver heating	Drive may become unstable	Load will be reduced and motor stall will be avoided.
Serial communication error	Wrong docking command	Simple command codes and safe stop logic will be used.

16. Future Scope

Future improvements may include:

- Guided-line navigation may be replaced with QR-code or AprilTag localization.
- A web dashboard may be added for route status and camera view.
- Battery monitoring may be added.

- Load-cell feedback may be added to measure product weight.
- Color classification may be upgraded to barcode or object recognition.
- Stronger motor drivers and a linear actuator may be used for heavier loads.
- An automatic charging dock may be added.

17. Sustainability and Reproducibility Compliance

17.1 Sustainability Commitments

- Modular parts will be used so sensors, motors, controllers, and battery can be replaced independently.
- The chassis, product samples, bins, and track materials will be reusable.
- Low-voltage DC operation and rechargeable battery power will be used.
- Local fabrication materials such as acrylic, PVC, plywood, or aluminum will be preferred.
- Unnecessary high-cost components will be avoided.

17.2 Reproducibility Commitments

- Commonly available components will be used.
- Open-source tools such as Python, OpenCV, Raspberry Pi OS, and Arduino IDE will be used.
- Wiring, pin allocation, and power distribution will be documented.
- HSV threshold values and line-following tuning values will be recorded.
- Each product color will be tested in repeated delivery trials.

18. Conclusion

This project will develop a vision-guided autonomous material handling robot that combines guided navigation, image-processing-based product classification, fork-lifter-based product handling, and autonomous docking selection. Although the robot will use a black-line route for reliable movement, the main autonomous feature will be its ability

to classify the product using computer vision and choose the correct destination without manual input.

The design will be realistic, affordable, and achievable within a student project timeline. By using Raspberry Pi for vision, Arduino Mega for real-time control, 25GA motors for movement, and a servo-driven fork-lifter for material handling, the prototype will demonstrate important concepts of mechatronics, robotics, industrial automation, and smart logistics.